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Understory response to springtime prescribed fire in two New York transition oak forests

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Abstract

Portions of two south-central New York transition oak stands received 0, 1 or 2 springtime prescribed fires between 1980/1981 and 1984. Observations 8–12 years later showed that forb richness, forb and shrub cover, and the importance values of forbs relative to shrubs increased in areas receiving one or two fires. *Fagus grandifolia, Dennstaedtia punctilobula, Gaultheria procumbens* and *Trientalis borealis* exhibited small, but detectable increases in importance values in burned areas. The importance values of *Quercus rubra, Kalmia latifolia* and *Viburnum acerifolium* decreased in burned areas. Areas receiving greater intensity fires contained taller tree regeneration and greater shrub cover, but no important changes in community composition could be attributed to fire intensity. An apparent relationship between rhizome depth and post-fire competitiveness of dominant perennial forbs and shrubs suggests the importance of below-ground morphological traits on survival and recovery following fire. Overall, the 12-year, post-burn cover and density of understory vegetation increased in burned areas and in areas receiving greater intensity fires, however, community composition remained largely unaffected by these springtime prescribed fires.

Keywords: Disturbance; Site preparation; *Quercus rubra*; Competitive strategies; Community composition

1. Introduction

Oaks (*Quercus*) were a major component of North American, eastern deciduous forests at the time of European colonization (Crow, 1988). Several studies have suggested that these species became established, and even maintained, by natural or/and anthropogenic fires (e.g. Brown, 1960; Lorimer, 1984; Abrams and Nowacki, 1992). Further, owing in part to disturbances accompanying expanding human settlements, oak species continue to dominate a substantial proportion of the eastern deciduous forest through modern times (Lorimer, 1984; Abrams, 1992; Abrams and Nowacki, 1992).

Despite their historic abundance, oak regeneration has more recently failed. Over major portions of their natural range, oaks are being replaced by more shade-tolerant, northern hardwood and mixed mesophytic species (e.g. Lorimer, 1984; Crow, 1988). The absence of fire in present-day oak forests may likely contribute to this transformation (Lorimer, 1985; Crow, 1988). However, field trials have not consistently demonstrated that fire, when used as a site preparation method, improves oak regeneration (Reich et al., 1990). Experimental results from Johnson (1974), Nyland et al. (1982), and Wendell and Smith

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(1986) show little benefit from using prescribed fire to promote oak establishment and development.

Different fire regimes (frequency, intensity, area) should yield distinct forest understories (i.e. shrubs and herbs) owing to different species' life histories. Yet, few studies (e.g. Buell and Cantlon, 1953; Little, 1974; Matlack and Good, 1989; DeSelm and Clebsch, 1991) have investigated effects of fire on understory vegetation of eastern deciduous forests. Before managers promote widespread use of prescribed fire as a site preparation technique for oaks, they must first understand its long-term effects upon associated species. Predictions of community responses to fire must consider species morphology (McLean, 1969; Flinn and Wein, 1977) and autecology (Gill, 1981; Trabaud, 1987), as well as the temporal and spatial variability in fire regime affecting the communities of interest (Cole et al., 1992; Rice, 1993).

The first objective of this study was to determine if prescribed springtime fires would improve the status of established, understory oak seedlings (advance regeneration), relative to the regeneration of competing canopy species. The second objective was to evaluate different frequencies (0-, 1- and 2-burns) and relative intensities (high and low) of prescribed fire for lasting (8–12 years) effects on understory structure and composition.

The following specific hypotheses were tested.

(1) More frequent and intense fires will favor the establishment and growth of disturbance-dependent and shade-intolerant tree species (e.g. some *Quercus*, *Betula* and *Populus* species).

(2) Different frequencies and intensities of prescribed fire will result in distinct understory communities.

2. Methods

2.1. Study site description

The two stands (Tompkins, 42°27′N, 76°20′W; Broome, 42°02′N, 75°26′W) used in this study were located within the oak–northern hardwood forest type of New York's Appalachian Highlands region (Stout, 1958). The study sites were selected for their abundant oak (*Quercus rubra*, *Quercus alba* L.) advance regeneration and their accessibility for fire control equipment (Adams, 1984).

The Tompkins site was located on the upper portion and crest of a west-facing slope, with elevations between 520 and 560 m. A 1955 aerial photograph (US Soil Conservation Service, 1965) shows the site located within a matrix of agricultural lands. Seral woodlots and old fields presently surround the site. A 1976 shelterwood cut removed approximately one-third of the stand basal area, leaving a residual 20 m^2 ha^{-1} in sawtimber-size trees (Adams, 1984), including 8.8 m^2 ha^{-1} of *Q. rubra* and *Q. alba* as large as 50–60 cm diameter at breast height (dbh). Table 1 lists the pre-fire (1979) overstory composition of the site.

Table 1
Pre-burn (1979) overstory composition (stems at least 9.0 cm dbh) of the Tompkins and Broome prescribed fire study sites, southern New York, USA[a,b] (mean ± SE)

Species	Density (stems ha^{-1})	Basal area (m^2 ha^{-1})
Tompkins		
Quercus rubra	76.21 ± 16.01	6.82 ± 1.58
Acer rubrum	243.00 ± 26.24	6.18 ± 0.74
Fagus grandifolia	102.94 ± 27.58	1.83 ± 0.49
Quercus alba	26.79 ± 7.86	1.98 ± 0.69
Populus grandidentata	16.46 ± 11.02	0.40 ± 0.25
Acer saccharum	12.36 ± 7.41	0.40 ± 0.25
Pinus strobus	12.36 ± 6.82	0.54 ± 0.44
Fraxinus americana	8.25 ± 4.84	0.59 ± 0.44
Tsuga canadensis	12.35 ± 6.13	0.44 ± 0.25
Others[c]	18.53 ± 4.99	0.69 ± 0.40
Total	529.25 ± 37.21	19.87 ± 1.48
Broome		
Quercus rubra	302.70 ± 39.04	16.36 ± 1.63
Acer rubrum	265.63 ± 30.00	5.68 ± 0.79
Fagus grandifolia	172.97 ± 36.18	3.21 ± 0.74
Betula lenta	74.13 ± 20.61	2.22 ± 0.64
Populus grandidentata	12.36 ± 5.39	0.69 ± 0.30
Amelanchier arborea	10.28 ± 4.20	0.10 ± 0.05
Others[d]	18.53 ± 5.83	0.69 ± 0.25
Total	856.60 ± 46.55	28.95 ± 1.33

[a] Modified from Adams (1984).
[b] Nomenclature follows Gleason and Cronquist (1991).
[c] Others include *Betula lenta*, *B. alleghaniensis*, *Q. velutina*, *Ostrya virginiana*, *Amelanchier arborea*, *Carya ovata*.
[d] Others include *Prunus serotina*, *T. canadensis*, *A. saccharum*, *Q. alba*, *P. strobus*.

The Broome site was located on the lower three-quarters of a southwest-facing slope, at elevations between 425 and 520 m. Regional foresters believe a salvage cut in the 1920s removed diseased American chestnut (*Castanea dentata* (Marsh.) Borkh.) from the site (Adams, 1984). No other records suggest later logging in this stand. Land in close proximity to the stand is entirely forested. Table 1 lists the pre-fire overstory composition.

Soils of the region are derived from glacial till, underlain by shale and sandstone. Soils at Tompkins are of the Lordstown and Volusia series (US Soil Conservation Service, 1965), and those at Broome of the Lordstown series (US Soil Conservation Service, 1971). Lordstown soils are coarse-loamy, mixed, mesic, Typic Dystrochrepts. They are well-drained and strongly acid, with depth to bedrock ranging from 50 to 100 cm. Sandstone outcrops are common on steeper, upper slopes. These were present at the Broome site. Volusia soils are deep, strongly acid, and somewhat poorly drained soils of medium texture, and are classified as fine-loamy, mixed, mesic, Aeric Fragiaquepts. A fragipan is usually present within 25–40 cm of the surface.

Both sites have a humid and continental climate. Average January maximum and minimum temperatures are respectively 0° and −9°C at Tompkins, and −1° and −8°C at Broome. Corresponding values for July are 27° and 14°C at Tompkins, and 25° and 15°C at Broome. Northeastern Tompkins County and eastern Broome County annually receive approximately 96 cm and 102 cm of precipitation. Precipitation in the region is uniformly distributed throughout the year (US Soil Conservation Service, 1965, 1971).

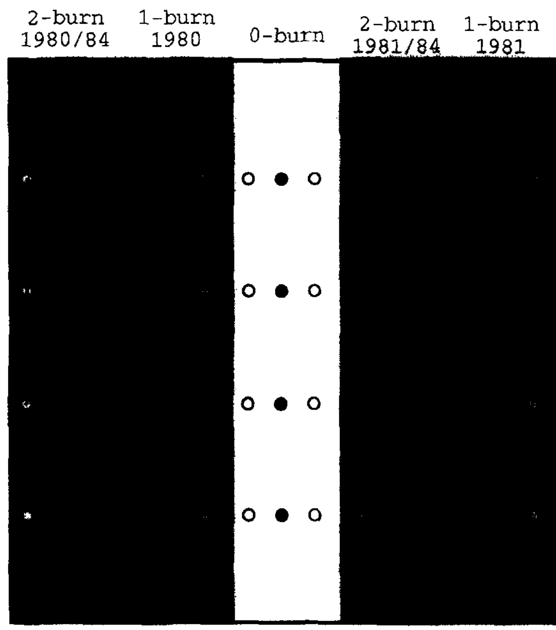

Fig. 1. General layout of the Tompkins and Broome prescribed fire study sites, southern New York, USA. The five treatment areas received one of three fire frequency treatments: 0-burn, 1-burn, or 2-burn. Nested permanent plot clusters are represented by open circles (original plots established in 1979) and filled circles (supplemental plots established in 1983). At Broome only, low intensity zones (light shading) were defined as areas within 15 m upslope or downslope of the rows of permanent plots. High intensity zones (dark shading) included areas beyond 15 m from permanent plot rows. Distances between rows of permanent plots ranged from 90 to 120 m. Randomly selected transect segments, used to collect data for fire intensity analyses at Broome, are illustrated as vertical lines.

2.2. Study site layout

Both sites were divided into five sections of approximately equal size, comprising three treatments: unburned, once-burned, and twice-burned (Fig. 1). The Tompkins and Broome sites were approximately 24 ha and 27 ha, respectively. In 1979, 1 year prior to the initial fires, nested permanent plot clusters were established in four rows running parallel to the slope contour, and crossing through all five treatment sections. Clusters included two 80 m^2 and three 4 m^2 plots. Each row originally contained two clusters per section. In 1983, one supplemental cluster was established in each section between the existing clusters (Fig. 1).

2.3. Burning techniques

The northern two sections of each site were burned in 1980 (the '1980 burn area'), and the southern sections in 1981 ('1981 burn area'). Each burn area was later subdivided and one-half reburned in April 1984. All burns were conducted by first establishing a backing fire (i.e. burns into wind or downhill) at the uphill boundary of each section. Flanking fires (i.e. burns parallel to wind or slope direction) were then set down the slope to the first row of permanent plots. A strip headfire (i.e. burns with wind or uphill) was then set across the slope, approximately 4 m downhill from the row of permanent plots. The headfires burned across the plots and continued uphill to meet the backing fire. Flanking fires and strip headfires were set sequentially down the slope, one sampling line at a time. A fifth strip headfire was set along the bottom boundary of each burn area.

2.4. Fire characteristics

Flame length, rate of spread, and fuel consumption were measured immediately adjacent to a few of the 4 m^2 permanent plots (Simard et al., 1982, 1984). Average (± 1 SD) fireline intensity during the Tompkins 1980 burn was 500 ± 237 kW m^{-1}, but intensities were not measured in 1981. The average fireline intensity during the Broome burns was 1997 ± 3187 kW m^{-1} in 1980, and 740 ± 495 kW m^{-1} in 1981. These data, and observations made during the burns, indicate that: fire intensity varied widely across each site; headfire intensities increased as the fire fronts advanced uphill; and fire intensities peaked where the headfires met the backing fires (Adams, 1984; R.D. Nyland, personal observations). Owing to high moisture content, ranging from 116 to 314%, the fires burned only the dried portions of the undecomposed leaf litter, leaving the humus unconsumed (Adams, 1984). Temperature on maximum recording thermometers placed at the soil–humus interface did not rise above air temperature during either the 1980 (Nyland et al., 1982) or 1981 fires (R.D. Nyland, unpublished data).

Soil chemistry analyses were conducted in August following the 1980 fires. At Tompkins, total first-year nitrogen, extractable phosphorus, and exchangeable potassium content (kg ha^{-1}) and concentration (percent dry weight) increased in the compacted litter. Nitrogen and phosphorus concentration increased in the humus (Nyland et al., 1982). At Broome, nitrogen, phosphorus and potassium content and concentration increased in the compacted litter. Potassium concentration increased in the humus. No changes were detected in the first-year nutrient content of the mineral soil at either site (Nyland et al., 1982).

2.5. Sampling methods

2.5.1. Fire frequency plot sampling

In 1979, tall woody stems (over 75 cm tall and less than 8.9 cm dbh) were tallied by height class on the permanent 80 m^2 plots. All forbs (with the exception of *Maianthemum canadense*, for which no quantitative data were taken in 1979) and short woody stems (less than or equal to 75 cm tall) were tallied by height class on the 4 m^2 plots. In 1992, herbaceous and woody stems less than 8.9 cm dbh were tallied by height class on all 4 m^2 plots. Sampling was conducted from June through August. Nomenclature follows Gleason and Cronquist (1991).

2.5.2. Fire intensity plot sampling

Observations made during the burns indicated that flame lengths and rates of spread generally increased with distance uphill from the firing lines. This suggested using distance from the firing lines as a surrogate for fire intensity. Areas within 15 m uphill or downhill from the permanent plots were classified as 'low intensity' zones. Those farther away were considered 'high intensity' zones (Fig. 1). Dry slash piles confounded the generalized fire intensity patterns at Tompkins. Therefore, fire intensity effects were examined only at Broome.

In 1992, at the Broome site, three transects were established in each burn area perpendicular to the rows of permanent plots at Broome (Fig. 1). Nested, rectangular 1 m^2 and 2 m^2 plots were established at 3 m intervals along the transects. Percent areal cover of all vegetation less than or equal to 1 m tall was estimated on the 1 m^2 plots using a modified Domin–Krajina cover class system (Mueller-Dombois and Ellenberg, 1974). Woody stems over 1 m tall and less than 8.9 cm dbh were tallied by height class on the 2 m^2 plots.

3. Analyses

For the purpose of statistical analyses, areas receiving the same fire frequency treatment were considered individual populations. Samples of response variables were taken in each population prior to and following treatment applications. Analyses of variance were conducted on the pre- and post-burn samples to evaluate changes in the response variables through time for each of the three populations at both study sites. Because of the lack of adequate replication of fire treatments at both sites ($N_{0\text{-burn}} = 1$, $N_{1\text{-burn}} = 2$, $N_{2\text{-burn}} = 2$), inferences drawn from the six populations apply only to those populations. Conclusions regarding fire effects (i.e. significant differences between pre- and post-treatment samples) must be tempered with the recognition that the effects of other ecological factors may have confounded the vegetation responses to fire in each of the areas.

3.1. Fire frequency effects

Analyses of variance were used to evaluate differences between pre-burn (1979) and post-burn (1992) levels of: species richness (number of species per 4 m^2 plot), density, cumulative height, and importance value (IV) of tree regeneration; and richness, cumulative height and importance value of understory species (herbs and shrubs). The variation among plots within each treatment was used for the ANOVA treatment error term. Species that never attained importance values of 0.05 or over were grouped as 'others'.

Response variables are defined as follows.

Density: number of stems m^{-2}.

Cumulative height:

$$\sum \mathrm{Density}_{\mathrm{HT}i} + \mathrm{Midpoint}_{\mathrm{HT}i}$$

where HTi is the ith height class, and $\mathrm{midpoint}_{\mathrm{HT}i}$ is the midpoint (cm) of the ith height class.

Importance value (IV):

$$IV = \frac{\text{Relative density} + \text{Relative cumulative height}}{2}$$

where

$$\text{Relative density} = \frac{\text{Species density}}{\text{Plot density}}$$

and where

$$\text{Relative cumulative height} = \frac{\text{Species cum. height}}{\text{Plot cum. height}}$$

The χ^2 goodness-of-fit statistic was used to test for significant differences in species frequencies (occurrence on 4 m^2 plots) between 1979 and 1992 for each of the three fire treatments.

For each study site, the 1979 and 1992 permanent plot data of species IV were divided into six groups representing the three fire treatment populations in the 2 sample years. Canonical discriminant function analyses were conducted to evaluate differences between the 1979 and 1992 species composition for each of the three fire treatment populations.

3.2. Fire intensity effects

Effects of fire intensity on tree regeneration and understory composition were analyzed using the 1992 transect data at Broome. The 2 m^2 plots yielded highly variable results; therefore, data from every three successive plots were combined to provide estimates of density and cover. In calculating cumulative height and importance values, percent cover was substituted for stem counts for individuals less than or equal to 100 cm tall (c.f. Eq. (1)):

Cumulative height

$$= \mathrm{Cover}_{\mathrm{HT}1} + \sum \mathrm{Midpoint}_{\mathrm{HT}i} \times \mathrm{Density}_{\mathrm{HT}i}$$

where HTi is the ith height class, and $\mathrm{Midpoint}_{\mathrm{HT}i}$ is the midpoint (cm) of the ith height class.

$$IV = \frac{\text{Species cumulative height}}{\text{Total cumulative height}}$$

Data from these 6 m^2 'combined plots' were used in analyses of variance to compare differences between high- and low-intensity fire zones within the 1- and 2-burn treatments for: species richness (number of species per 6 m^2 combined plot), cumulative height, and IV of tree regeneration; and richness, cumulative height and IV of understory species. Between plot variation within each treatment was used for the ANOVA treatment error term.

The χ^2 goodness-of-fit statistic was used to test for significant differences in species frequencies (occurrence on 6 m^2 combined plots) between high- and low-intensity zones in each fire treatment.

The data set of species IV on each combined plot was divided into four groups representing the two fire intensity levels in the 1- and 2-burn areas. A canonical discriminant function analysis was conducted to evaluate differences in overall species composition between the high- and low-intensity zones of both burn frequency areas.

Table 2
Study site vascular plant species richness of pre-burn (1979) and 12-year post-burn (1992) flora at the Tompkins and Broome prescribed fire study sites, southern New York, USA. Values are the total number of species observed on original 4 m^2 plots in 1979 and 1992 in areas receiving different fire treatments. The sample size (n) for each treatment area is provided

	0-burn		1-burn		2-burn	
	1979	1992	1979	1992	1979	1992
Tompkins						
n	24		54		48	
Forbs	20	28	31	32	26	44
Shrubs	6	6	8	6	6	7
Trees	16	17	18	18	15	17
Broome						
n	27		48		51	
Forbs	15	16	22	28	23	27
Shrubs	2	3	7	8	8	5
Trees	9	10	15	16	18	17

4. Results

4.1. Fire frequency effects

4.1.1. Species richness

The richness of the pre-burn and 12-year post-burn tree and shrub flora remained relatively constant at both sites (Table 2). Forb richness at Tompkins exhibited an inconsistent response to fire treatments. Richness increased 1.4-fold in the 0-burn areas, 1.7-fold in the 2-burn areas, and only nominally in the 1-burn areas. At Broome, forb richness increased 1.25- and 1.2-fold in the 1- and 2-burn areas, but only nominally in the 0-burn area. Table 3 compares pre- and post-burn species richness on a per plot basis. At both study sites, the number of forb species per 4 m^2 plot increased between 1979 and 1992 in both burn areas, but not in the 0-burn area. The magnitude of these increases ranged from 1.2 to 1.9-fold. The number of shrub species per 4 m^2 plot remained unchanged in all treatment areas at both sites. There were no discernible changes in tree richness at Broome, but at Tompkins, the 4 m^2 plots averaged one fewer species in 1992 across all treatments.

4.1.2. Cumulative height

At Tompkins, total forb and shrub cumulative heights increased between 1979 and 1992 by 4.1- and 1.9-fold, respectively, in the 2-burn areas (Table 4). No significant changes in forb and shrub cumulative height were found in the 0-burn or 1-burn areas. Likewise, no changes could be detected for the cumulative height of total tree regeneration in any treatment area. Species-specific changes in cumulative heights are indicated in Table 4.

Total tree regeneration density (not reported in Table 4) decreased significantly from 12.7 ± 1.4 to 5.6 ± 0.5 stems m^{-2} in the 1-burn, and 14.3 ± 1.6 to 5.6 ± 0.4 stems m^{-2} in the 2-burn areas ($P < 0.01$ for both burn areas). Tree regeneration density remained unchanged in the 0-burn area (density was

Table 3
Permanent plot vascular plant species richness (mean ± SE number of species per 4 m^2 plot) at the Tompkins and Broome prescribed fire study sites, southern New York, USA. Pre-burn (1979) and post-burn (1992) values are paired by fire frequency treatment. Significant differences between paired values are denoted by: * $P \leq 0.05$, ** $P \leq 0.01$. The sample size (n) for each treatment is provided

Growth form	0-burn		1-burn		2-burn	
	1979	1992	1979	1992	1979	1992
Tompkins						
n	24	36	54	78	48	72
Forbs	4.5 ± 0.4	4.9 ± 0.4	2.8 ± 0.2	3.6 ± 0.2 *	2.3 ± 0.2	4.3 ± 0.3 **
Shrubs	2.0 ± 0.2	2.0 ± 0.2	1.4 ± 0.2	1.3 ± 0.1	1.1 ± 0.1	1.3 ± 0.1
Trees	6.2 ± 0.5	5.3 ± 0.4 *	5.7 ± 0.3	4.5 ± 0.3 **	5.3 ± 0.2	4.1 ± 0.2 **
Broome						
n	27	36	48	66	51	69
Forbs	4.2 ± 0.5	4.1 ± 0.4	4.7 ± 0.3	5.6 ± 0.3 *	4.6 ± 0.3	6.0 ± 0.3 **
Shrubs	1.0 ± 0.1	1.0 ± 0.1	1.2 ± 0.1	1.1 ± 0.1	1.1 ± 0.1	0.9 ± 0.1
Trees	2.7 ± 0.3	3.1 ± 0.2	3.9 ± 0.3	3.9 ± 0.2	4.1 ± 0.3	4.5 ± 0.2

Table 4

Summary of understory and tree regeneration structure and composition in respective fire frequency treatments at the Tompkins and Broome prescribed fire study sites, southern New York, USA. Values listed are for mean ± 1S.E. cumulative height (cm growth/m^{-2}), frequency of occurrence on 4 m^2 plots, and importance values (IV = (relative density + relative cumulative height)/2). See Analyses for details of cumulative height and importance value calculations. Pre-burn (1979) and post-burn (1992) values are paired by fire frequency treatment. Significant differences between paired values are denoted by: * $P \leq 0.05$, ** $P \leq 0.01$. The sample size (n) for each treatment is provided

	0-burn		1-Burn		2-Burn	
	1979 (n = 16)	1992 (n = 36)	1979 (n = 36)	1992 (n = 78)	1979 (n = 32)	1992 (n = 72)
Tompkins						
Cumulative height						
Dennstaedtia punctilobula	0.0 ± 0.0	0.1 ± 0.1	2.3 ± 1.4	22.3 ± 8.3	7.3 ± 6.8	77.8 ± 30.1 *
Gaultheria procumbens	1.3 ± 0.7	4.5 ± 1.4	0.4 ± 0.2	1.7 ± 0.5	0.3 ± 0.1	1.7 ± 1.2
Mitchella repens	0.3 ± 0.2	4.7 ± 1.9 **	0.2 ± 0.1	2.4 ± 0.7	0.4 ± 0.2	1.2 ± 0.4
Trientalis borealis	9.6 ± 3.6	8.8 ± 1.8	5.3 ± 2.1	5.0 ± 1.0	3.8 ± 2.2	11.5 ± 3.1 *
Uvularia sessilifolia	8.1 ± 5.4	13.2 ± 5.6	11.3 ± 6.1	5.3 ± 1.3	5.0 ± 2.2	33.3 ± 13.4 *
Other forbs	39.5 ± 22.5	27.4 ± 8.5	22.5 ± 7.9	50.6 ± 8.7	40.1 ± 26.2	109.4 ± 26.1 **
Total forbs	58.9 ± 17.7	58.7 ± 13.6	42.0 ± 9.2	87.3 ± 13.8	56.9 ± 27.9	236.0 ± 40.4 **
Rhododendron periclymenoides	99.7 ± 29.1	64.2 ± 19.2	33.3 ± 17.1	37.8 ± 10.7	13.0 ± 6.3	3.8 ± 2.3
Rubus spp. [a]	3.4 ± 2.9	4.6 ± 2.7	58.1 ± 21.4	33.0 ± 15.5	57.0 ± 22.4	160.2 ± 41.0 **
Vaccinium spp.	21.0 ± 8.1	16.6 ± 4.3	7.1 ± 5.2	19.5 ± 7.9	1.2 ± 1.0	2.0 ± 1.0
Viburnum acerifolium	43.9 ± 33.5	49.4 ± 11.6	41.6 ± 10.3	62.1 ± 12.1	47.2 ± 13.4	51.5 ± 15.5
Other shrubs	0.0 ± 0.0	0.2 ± 0.1	1.5 ± 1.3	1.4 ± 0.6	0.2 ± 0.2	5.4 ± 2.4
Total shrubs	168.1 ± 43.4	134.9 ± 24.4	141.6 ± 27.8	153.8 ± 27.1	118.5 ± 25.6	222.7 ± 43.0 **
Total understory	226.9 ± 41.2	193.6 ± 27.4	183.7 ± 29.1	241.1 ± 30.4	175.5 ± 38.8	458.7 ± 62.5 **
Acer rubrum	36.5 ± 5.5	7.9 ± 3.9	62.7 ± 20.1	42.7 ± 10.8	44.5 ± 7.3	38.3 ± 10.4
Amelanchier arborea	12.4 ± 4.1	71.7 ± 22.7 **	6.8 ± 2.9	21.6 ± 6.4	3.5 ± 2.1	13.0 ± 5.3
Fagus grandifolia	66.8 ± 12.7	103.5 ± 20.7	70.0 ± 11.0	176.5 ± 27.6 **	84.6 ± 22.8	152.7 ± 27.6
Fraxinus americana	6.1 ± 2.6	12.3 ± 6.2	29.0 ± 8.6	27.0 ± 8.4	35.6 ± 12.4	12.7 ± 3.2 *
Hamamelis virginiana	27.5 ± 10.0	60.3 ± 18.3	22.3 ± 6.1	22.1 ± 6.9	18.3 ± 5.9	24.3 ± 9.7
Quercus rubra	42.6 ± 8.4	8.4 ± 4.5	46.4 ± 15.4	22.9 ± 6.1	64.9 ± 21.8	31.9 ± 9.3 *
Other trees	93.9 ± 22.7	85.7 ± 23.7	80.6 ± 13.5	81.0 ± 13.0	78.7 ± 15.1	86.0 ± 18.0
Total trees	285.9 ± 28.9	425.1 ± 39.0	317.7 ± 31.0	393.9 ± 34.2	330.1 ± 38.8	358.8 ± 35.5
Frequency						
Dennstaedtia punctilobula	0.00	0.04	0.07	0.13	0.06	0.23 *
Gaultheria procumbens	0.33	0.58	0.20	0.30	0.17	0.23
Maianthemum canadense [b]	0.50	0.71	0.31	0.63 **	0.21	0.35
Mitchella repens	0.21	0.46	0.24	0.41	0.19	0.25
Trientalis borealis	0.17	0.67	0.37	0.44	0.27	0.26
Uvularia sessilifolia	0.38	0.42	0.28	0.30	0.33	0.42
Other forbs	0.83	0.75	0.69	0.93	0.52	0.67
Rhododendron periclymenoides	0.63	0.58	0.26	0.30	0.19	0.17
Rubus spp. [a]	0.17	0.12	0.31	0.24	0.31	0.44
Vaccinium spp.	0.75	0.58	0.26	0.45	0.08	0.10
Viburnum acerifolium	0.50	0.54	0.56	0.52	0.48	0.44
Other shrubs	0.04	0.08	0.07	0.07	0.02	0.10
Acer rubrum	0.96	0.92	0.93	0.78	0.92	0.71
Amelanchier arborea	0.46	0.58	0.20	0.46 **	0.08	0.23 *
Fagus grandifolia	0.63	0.75	0.67	0.80	0.69	0.67
Fraxinus americana	0.46	0.21	0.59	0.35 **	0.65	0.46
Hamamelis virginiana	0.50	0.58	0.37	0.33	0.21	0.27
Quercus rubra	0.88	0.38 **	0.80	0.37 **	0.88	0.54 **
Other trees	0.92	0.79	0.87	0.74	0.90	0.58 **

10.1 ± 1.1 stems m^{-2} in 1979 and 7.6 ± 1.0 stems m^{-2} in 1992).

At Broome, the total cumulative height of forbs increased 1.7- and 2-fold in the 1- and 2-burn areas, and were unchanged in the unburned area. Cumulative heights of shrubs were unchanged in all treat-

Table 4 (continued)

	0-burn		1-Burn		2-Burn	
	1979 ($n = 16$)	1992 ($n = 36$)	1979 ($n = 36$)	1992 ($n = 78$)	1979 ($n = 32$)	1992 ($n = 72$)
Importance value						
Dennstaedtia punctilobula	0.00 ± 0.00	0.01 ± 0.01	0.01 ± 0.01	0.07 ± 0.02	0.02 ± 0.02	0.09 ± 0.02 *
Gaultheria procumbens	0.08 ± 0.06	0.09 ± 0.03	0.02 ± 0.02	0.04 ± 0.02	0.03 ± 0.02	0.03 ± 0.02
Mitchella repens	0.01 ± 0.01	0.07 ± 0.03 *	0.03 ± 0.02	0.06 ± 0.02	0.03 ± 0.02	0.03 ± 0.01
Trientalis borealis	0.05 ± 0.02	0.07 ± 0.02	0.04 ± 0.01	0.05 ± 0.01	0.06 ± 0.03	0.10 ± 0.02
Uvularia sessilifolia	0.09 ± 0.06	0.10 ± 0.04	0.13 ± 0.05	0.10 ± 0.02	0.12 ± 0.05	0.14 ± 0.03
Other forbs	0.17 ± 0.05	0.14 ± 0.03	0.20 ± 0.05	0.25 ± 0.04	0.15 ± 0.04	0.23 ± 0.03
Total forbs	0.39 ± 0.10	0.50 ± 0.05	0.43 ± 0.06	0.56 ± 0.04	0.41 ± 0.06	0.61 ± 0.04 **
Rhododendron periclymenoides	0.35 ± 0.08	0.19 ± 0.04 **	0.11 ± 0.04	0.07 ± 0.02	0.07 ± 0.04	0.02 ± 0.01
Rubus spp. [a]	0.03 ± 0.02	0.01 ± 0.01	0.17 ± 0.06	0.07 ± 0.02	0.15 ± 0.05	0.18 ± 0.03
Vaccinium spp.	0.08 ± 0.03	0.08 ± 0.02	0.03 ± 0.02	0.05 ± 0.02	0.03 ± 0.02	0.01 ± 0.01
Viburnum. acerifolium	0.08 ± 0.05	0.21 ± 0.05	0.22 ± 0.04	0.18 ± 0.03	0.26 ± 0.06	0.15 ± 0.03 *
Other shrubs	0.00 ± 0.00	0.01 ± 0.01	0.02 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01
Total shrubs	0.55 ± 0.10	0.50 ± 0.05	0.55 ± 0.06	0.38 ± 0.04 *	0.52 ± 0.07	0.37 ± 0.04 *
Acer rubrum	0.18 ± 0.03	0.30 ± 0.03 *	0.20 ± 0.03	0.16 ± 0.02	0.22 ± 0.03	0.15 ± 0.02
Amelanchier arborea	0.04 ± 0.02	0.16 ± 0.04 **	0.02 ± 0.01	0.07 ± 0.02 *	0.01 ± 0.01	0.04 ± 0.01
Fagus grandifolia	0.20 ± 0.04	0.22 ± 0.04	0.20 ± 0.03	0.35 ± 0.04 **	0.21 ± 0.04	0.35 ± 0.04 *
Fraxinus americana	0.03 ± 0.01	0.03 ± 0.02	0.12 ± 0.03	0.11 ± 0.02	0.13 ± 0.03	0.07 ± 0.02
Hamamelis virginiana	0.07 ± 0.02	0.10 ± 0.03	0.05 ± 0.01	0.05 ± 0.01	0.04 ± 0.01	0.05 ± 0.01
Quercus rubra	0.19 ± 0.03	0.03 ± 0.01 **	0.15 ± 0.05	0.06 ± 0.01 **	0.20 ± 0.03	0.10 ± 0.02 **
Other trees	0.29 ± 0.05	0.17 ± 0.02	0.25 ± 0.03	0.20 ± 0.02	0.20 ± 0.03	0.25 ± 0.03
Broome						
Cumulative height						
Aralia nudicaulis	35.8 ± 14.1	19.8 ± 6.1	23.1 ± 7.7	29.2 ± 6.1	40.5 ± 11.9	39.1 ± 10.3
Gaultheria procumbens	2.0 ± 0.7	3.7 ± 1.0	3.3 ± 0.8	10.4 ± 2.1 **	2.4 ± 0.6	13.0 ± 2.2 **
Mitchella repens	0.9 ± 0.5	2.9 ± 0.7	1.2 ± 0.3	3.1 ± 0.7 **	1.2 ± 0.3	3.6 ± 0.7 **
Trientalis borealis	10.7 ± 3.5	7.3 ± 2.7	9.7 ± 1.8	21.4 ± 4.0 *	5.9 ± 1.4	29.0 ± 3.4 **
Uvularia sessilifolia	4.9 ± 2.3	6.3 ± 2.3	5.3 ± 2.0	16.4 ± 4.7	3.6 ± 1.7	33.4 ± 11.5 **
Other forbs	18.2 ± 10.3	9.9 ± 3.2	36.7 ± 15.2	56.2 ± 15.5	33.0 ± 9.5	56.2 ± 10.0
Total forbs	72.5 ± 22.0	49.9 ± 9.1	79.4 ± 17.5	136.8 ± 19.3 *	86.6 ± 13.2	174.4 ± 22.0 **
Kalmia latifolia	95.6 ± 35.0	75.2 ± 18.4	72.0 ± 19.2	89.1 ± 22.3	75.1 ± 19.2	66.4 ± 16.7
Viburnum acerifolium	25.8 ± 11.0	19.2 ± 30.4	27.8 ± 7.8	30.4 ± 7.1	32.8 ± 11.7	29.3 ± 9.2
Other shrubs	0.3 ± 0.3	1.2 ± 1.2	1.3 ± 1.0	6.8 ± 2.9	4.0 ± 3.1	25.6 ± 18.2
Total shrubs	121.7 ± 39.1	95.6 ± 20.4	101.1 ± 19.0	126.2 ± 22.5	112.0 ± 19.8	121.3 ± 25.3
Total understory	194.2 ± 40.6	145.5 ± 22.0	180.5 ± 26.4	263.0 ± 32.1 *	198.5 ± 27.6	295.7 ± 34.9 *
Acer rubrum	12.9 ± 2.3	13.1 ± 3.9	17.6 ± 3.0	25.4 ± 4.6 *	28.7 ± 4.7	35.2 ± 3.8
Amelanchier arborea	7.7 ± 4.0	7.3 ± 6.3	11.9 ± 3.4	15.6 ± 5.1	17.3 ± 3.5	21.8 ± 3.4
Fagus grandifolia	59.9 ± 13.1	95.6 ± 22.6	53.0 ± 7.2	172.0 ± 26.8 *	69.7 ± 11.2	170.1 ± 37.9 **
Hamamelis virginiana	23.5 ± 4.7	20.1 ± 15.5	31.0 ± 9.5	21.5 ± 8.1	26.0 ± 10.0	25.4 ± 9.5
Quercus rubra	29.9 ± 10.4	9.9 ± 1.9	45.6 ± 8.1	14.8 ± 3.0 *	66.7 ± 16.9	46.8 ± 10.2
Other trees	23.9 ± 8.4	16.0 ± 6.4	36.4 ± 8.3	38.2 ± 15.9	41.4 ± 6.5	37.7 ± 9.4.
Total trees	157.8 ± 16.4	161.9 ± 29.7	195.5 ± 25.0	287.4 ± 32.0	249.9 ± 21.9	337.0 ± 37.2 *

ment areas at Broome. Total tree cumulative height increased only in the 2-burn areas, by 1.3-fold. Total tree cumulative height did not change in 1-burn areas, although differences occurred among some individual species (Table 4).

At Broome, mean tree regeneration densities were unchanged between 1979 and 1992 in all treatments. Mean density ranged from 2.3 to 7.8 stems m^{-2}, values which were consistently lower than those at Tompkins.

Table 4 (continued)

	0-burn		1-Burn		2-Burn	
	1979 ($n = 16$)	1992 ($n = 36$)	1979 ($n = 36$)	1992 ($n = 78$)	1979 ($n = 32$)	1992 ($n = 72$)
Frequency						
Aralia nudicaulis	0.44	0.33	0.54	0.50	0.41	0.33
Gaultheria procumbens	0.59	0.52	0.60	0.71	0.59	0.67
Maianthemum canadense [b]	0.44	0.48	0.48	0.63 *	0.45	0.75 **
Mitchella repens	0.37	0.52	0.52	0.52	0.53	0.61
Trientalis borealis	0.67	0.44	0.67	0.90 **	0.55	0.82 **
Uvularia sessilifolia	0.22	0.30	0.33	0.31	0.22	0.41 *
Other forbs	0.67	0.59	0.58	0.79 *	0.76	0.84
Kalmia latifolia	0.56	0.48	0.54	0.45	0.51	0.43
Viburnum acerifolium	0.48	0.44	0.42	0.42	0.41	0.41
Other shrubs	0.00	0.04	0.19	0.15	0.16	0.06
Acer rubrum	0.74	0.74	0.73	0.85	0.78	0.88
Amelanchier arborea	0.22	0.19	0.48	0.50	0.55	0.71
Fagus grandifolia	0.63	0.74	0.50	0.83 **	0.53	0.78 **
Hamamelis virginiana	0.37	0.33	0.40	0.40	0.33	0.35
Quercus rubra	0.48	0.67	0.81	0.65 **	0.78	0.71
Other trees	0.30	0.37	0.50	0.58	0.65	0.58
Importance value						
Aralia nudicaulis	0.17 ± 0.07	0.11 ± 0.03	0.10 ± 0.03	0.08 ± 0.02	0.12 ± 0.03	0.08 ± 0.02
Gaultheria procumbens	0.08 ± 0.02	0.09 ± 0.02	0.10 ± 0.02	0.13 ± 0.02	0.09 ± 0.02	0.17 ± 0.03 **
Mitchella repens	0.07 ± 0.06	0.10 ± 0.03	0.04 ± 0.01	0.07 ± 0.02	0.05 ± 0.02	0.06 ± 0.01
Trientalis borealis	0.14 ± 0.04	0.07 ± 0.02	0.17 ± 0.04	0.20 ± 0.03	0.09 ± 0.02	0.17 ± 0.02 *
Uvularia sessilifolia	0.09 ± 0.05	0.10 ± 0.03	0.05 ± 0.02	0.08 ± 0.02	0.05 ± 0.02	0.10 ± 0.02
Other forbs	0.07 ± 0.03	0.11 ± 0.03	0.14 ± 0.03	0.16 ± 0.03	0.20 ± 0.05	0.20 ± 0.03
Total forbs	0.61 ± 0.09	0.58 ± 0.06	0.60 ± 0.06	0.73 ± 0.03 *	0.59 ± 0.05	0.79 ± 0.03 **
Kalmia latifolia	0.30 ± 0.09	0.32 ± 0.06	0.24 ± 0.06	0.15 ± 0.03	0.20 ± 0.05	0.11 ± 0.02 *
Viburnum acerifolium	0.09 ± 0.03	0.08 ± 0.02	0.15 ± 0.05	0.10 ± 0.02	0.15 ± 0.04	0.08 ± 0.02 *
Other shrubs	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.02 ± 0.01	0.02 ± 0.01	0.03 ± 0.01
Total shrubs	0.39 ± 0.09	0.40 ± 0.06	0.40 ± 0.06	0.27 ± 0.03 *	0.38 ± 0.05	0.21 ± 0.03 **
Acer rubrum	0.19 ± 0.04	0.19 ± 0.03	0.19 ± 0.03	0.21 ± 0.03	0.19 ± 0.02	0.21 ± 0.02
Amelanchier arborea	0.06 ± 0.01	0.04 ± 0.02	0.06 ± 0.02	0.07 ± 0.02	0.07 ± 0.02	0.09 ± 0.01
Fagus grandifolia	0.31 ± 0.06	0.39 ± 0.06	0.25 ± 0.04	0.43 ± 0.04 *	0.26 ± 0.05	0.34 ± 0.04
Hamamelis virginiana	0.15 ± 0.05	0.07 ± 0.02	0.12 ± 0.03	0.07 ± 0.02 *	0.07 ± 0.02	0.06 ± 0.02
Quercus rubra	0.20 ± 0.06	0.19 ± 0.04	0.23 ± 0.04	0.12 ± 0.02 **	0.26 ± 0.04	0.19 ± 0.03 *
Other trees	0.09 ± 0.02	0.10 ± 0.03	0.15 ± 0.03	0.10 ± 0.02	0.15 ± 0.02	0.10 ± 0.02

[a] *Rubus* spp. includes primarily *R. allegheniensis*, but also *R. idaeus*.

[b] No quantitative pre-burn sampling was conducted for *M. canadense*. Therefore, 1979 cumulative heights and importance values could not be calculated for this species. Data are presented for frequency.

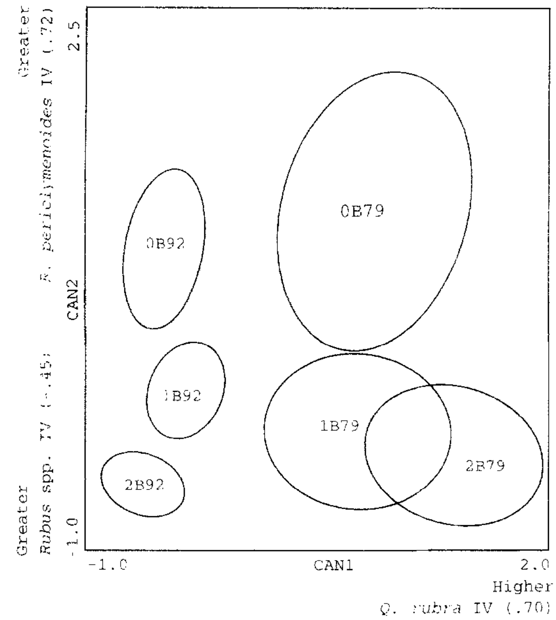

Fig. 2. Canonical discriminant function analysis of species importance values (IV) at the Tompkins prescribed fire study site, southern New York, USA. The six groups represent the pre-burn (1979) and post-burn (1992) species composition in the 0-burn, 1-burn and 2-burn treatment areas. Within-group correlations between species and the canonical axes are provided. Groups are represented by 95% confidence ellipses for the bivariate mean of the first two canonical axes (Sokal and Rohlf, 1981). The number of samples (n) included in each group are: 0B79, 16; 0B92, 36; 1B79, 36; 1B92, 78; 2B79, 32; 2B92, 72.

4.1.3. Importance values

At Tompkins, the overall IV of forbs increased relative to shrubs in both burned areas. Forb IV increased 1.5-fold in the 2-burn areas between 1979 and 1992, while that for shrubs decreased by a factor of approximately 0.7 in both the 1- and 2-burn areas (Table 4). The discriminant analysis successfully separated the 1992 plots from the 1979 plots in each of the three treatment areas (Fig. 2). The 1992 and 1979 plots separated along the first canonical axis, which accounted for 51% of the total variation, and had a canonical correlation of 0.59 ($P < 0.001$) with the grouping variable. This axis represents *Q. rubra* IV gradient (within group correlation of 0.70). All other species had low (less than 0.40) correlations with this axis.

These results are consistent with the ANOVA results (Table 4). *Quercus rubra IV* decreased more than 0.5-fold between 1979 and 1992 in all treatment areas owing to lower frequencies in all treatments and smaller cumulative heights in the 2-burn treatment in 1992, compared with 1979. ANOVA detected additional IV differences. *Mitchella repens IV* increased in the 0-burn area owing to an increased cumulative height there. *Rhododendron periclymenoides* decreased in the 0-burn area. In 2-burn areas, the IV of *Dennstaedtia punctilobula* increased, and that of *Viburnum acerifolium* decreased. Although the cumulative heights of *Trientalis borealis, Uvularia sessilifolia* and *Rubus* spp. all increased in the 2-burn areas, their IV did not significantly change. The IV of *F. grandifolia* increased in both burn areas, reflecting greater cumulative heights and frequencies.

The second canonical axis (canonical correlation of 0.49, $P < 0.001$) accounted for 30% of the total variation, and represents a gradient contrasting *Rhododendron periclymendoides* and *Rubus* spp. importance values (within group correlations of 0.72 and -0.45, respectively). This axis separated the 0-burn area from the burned areas, reflecting the

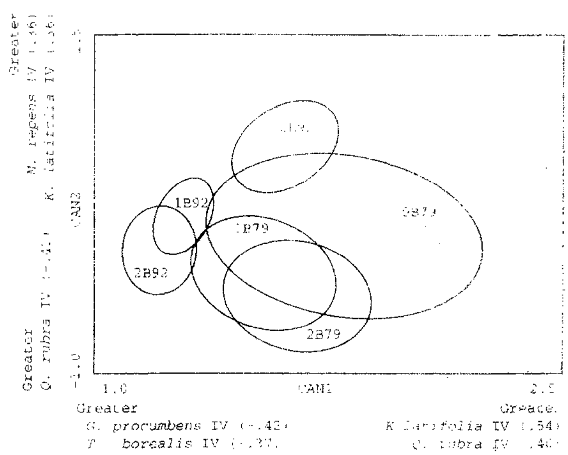

Fig. 3. Canonical discriminant function analysis of species importance values (IV) at the Broome prescribed fire study site, southern New York, USA. The six groups represent the pre-burn (1979) and post-burn (1992) species composition in the 0-burn, 1-burn and 2-burn treatment areas. Within-group correlations between species and the canonical axes are provided. Groups are represented by 95% confidence ellipses for the bivariate mean of the first two canonical axes (Sokal and Rohlf, 1981). The number of samples (n) included in each group are: 0B79, 18; 0B92, 36; 1B79, 32; 1B92, 66; 2B79, 34; 2B92, 69.

Table 5
Mean ± SE species richness (number of species per 6 m^2 combined plot) in high- and low-intensity zones of the 1- and 2-burn areas at the Broome prescribed fire study site, southern New York, USA. Values are paired by fire frequency treatment. Significant differences between paired values are denoted by: * $P \leq 0.05$, ** $P \leq 0.01$. The sample size (n) for each treatment is provided

Growth form	1-burn		2-burn	
	High ($n=73$)	Low ($n=59$)	High ($n=93$)	Low ($n=66$)
Graminoids	0.2±0.1	0.2±0.1	0.6±0.1	0.5±0.1
Forbs	6.1±0.4	6.1±0.4	5.8±0.7	4.3±0.3
Shrubs	1.2±0.1	1.3±0.1	1.5±0.1	0.8±0.1 **
Trees	4.8±0.2	4.2±0.2	4.9±0.2	4.1±0.2 **
Total	12.2±0.6	11.8±0.1	12.8±0.5	10.7±0.5 **

overall greater importance of *Rhododendron periclymenoides* in the 0-burn area. However, this condition existed in 1979, prior to the fires (Table 4). Similarly, the second axis reflects the greater importance of *Rubus* spp. in the burned areas prior to the fires, as well as the persistence of *Rubus* spp. in the 2-burn areas after the fires (Table 4).

At Broome the *IV* for all growth forms and species remained unchanged in the 0-burn area (Table 4). The importance of forbs increased 1.2-fold in the 1-burn areas and 1.3-fold in the 2-burn areas, while that of shrubs decreased 0.7-fold in the 1-burn and 0.6-fold in the 2-burn areas.

Changes in species *IV*, occurring between 1979 and 1992 in the burned areas, are reflected in the results of the discriminant analysis (Fig. 3). The first canonical axis accounted for 45% of the total variation, and had a canonical correlation of 0.41 ($P < 0.001$) with the grouping variable. This axis distinguishes the 1992 vegetation from that of 1979 in both burned areas. It represents a gradient of lower *Kalmia latifolia* and *Q. rubra IV*, and greater *Gaultheria procumbens* and *T. borealis IV* in the burned areas in 1992 (respective within group correlations: 0.54, 0.40, −0.42 and −0.37). The second axis accounted for 24% of the total variation and had a canonical correlation of 0.31 ($P = 0.06$). This axis does not serve to distinguish between pre- and post-treatment vegetation in any of the treatment areas. It does illustrate the fact that a pre-burn *IV* gradient existed between the treatment areas for *Mitchella repens*, *K. latifolia* and *Q. rubra* (respective within group correlations: 0.36, 0.36 and −0.47). This gradient, which simply reflects an unequal distribution of these species across the study site in 1979, still remained in 1992.

Results of the discriminant analysis are consistent with the ANOVA results (Table 4): *Q. rubra IV* remained unchanged in the 0-burn area, but decreased by a factor of 0.5 in the 1-burn area and 0.7 in the 2-burn area, reflecting its lower frequency and cumulative height following the fires; *K. latifolia IV* decreased by a factor of 0.6 in the 2-burn areas; and even though the cumulative heights of *G. procumbens*, *Mitchella repens*, *T. borealis* and *U. sessilifolia* all increased in both burn treatments, only *G. procumbens* and *T. borealis* had higher *IV*s in the 2-burn areas. ANOVA detected additional *IV* differences: *Hamamelis virginiana IV* decreased and *F. grandifolia* increased by factors of 0.6 and 1.7, respectively, in the 1-burn areas.

4.2. Fire intensity effects

4.2.1. Species richness

The high-intensity zones (over 15 m from firing line) of the 2-burn areas had 1.2- and 1.9-fold higher

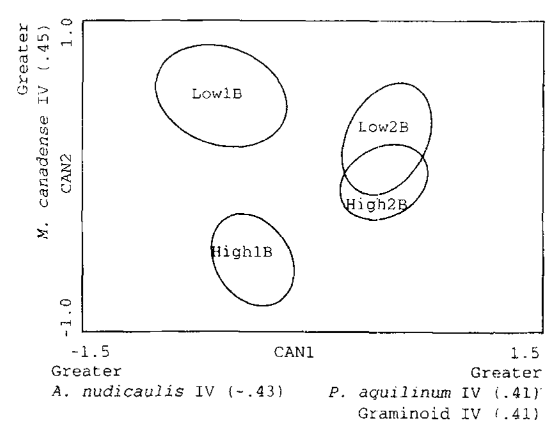

Fig. 4. Canonical discriminant function analysis of species importance values (*IV*) in the high- and low-intensity zones at the Broome prescribed fire study site, southern New York, USA. The four groups represent the post-burn (1992) species composition in the high- and low-intensity zones of the 1-burn and 2-burn treatment areas. Within-group correlations between species and the canonical axes are provided. Groups are represented by 95% confidence ellipses for the bivariate mean of the first two canonical axes (Sokal and Rohlf, 1981). The number of samples (n) included in each group are: LOW1B, 59; HIGH1B, 73; LOW2B, 66; HIGH2B, 93.

Table 6
Summary of 1992 understory and tree regeneration structure and composition in fire intensity zones at the Broome prescribed fire study site, southern New York, USA. Values presented are for mean ± 1 SE cumulative height (% cover for stems ≤ 100 cm) + (cm growth for stems > 100 cm)/m^2), frequency of occurrence on 6 m^2 combined plots, and mean ± 1 SE importance value (relative cumulative height). See Analyses for details of cumulative height and importance value calculations. Values for high- and low-intensity zones are paired by fire frequency treatment. Significant differences between paired values are denoted by: * $P \le 0.05$, ** $P \le 0.01$. The sample size (n) for each treatment is provided

	1-burn		2-burn	
	High (n = 73)	Low (n = 59)	High (n = 93)	Low (n = 66)
Cumulative height				
Total graminoid	0.1 ± 0.1	0.2 ± 0.1	0.8 ± 0.2	0.6 ± 0.2
Aralia nudicaulis	3.7 ± 0.8	6.8 ± 1.2 **	2.4 ± 0.5	3.2 ± 1.0
Dennstaedtia punctilobula	5.3 ± 0.3	2.5 ± 0.9 *	2.0 ± 0.8	1.9 ± 0.7
Gaultheria procumbens	0.3 ± 0.1	0.2 ± 0.1	0.5 ± 0.1	0.3 ± 0.1 *
Maianthemum canadense	0.2 ± 0.1	0.4 ± 0.1 **	0.3 ± 0.1	0.4 ± 0.1
Mitchella repens	0.1 ± 0.1	0.2 ± 0.1	0.2 ± 0.1	0.1 ± 0.1
Pteridium aquilinum	1.4 ± 0.5	1.1 ± 0.4	4.6 ± 1.0	2.6 ± 0.9 *
Trientalis borealis	0.2 ± 0.1	0.3 ± 0.1 **	0.3 ± 0.2	0.3 ± 0.1
Other forb	1.4 ± 0.3	2.1 ± 0.8	1.0 ± 0.2	0.5 ± 0.1
Total forb	12.7 ± 1.6	13.5 ± 1.7	11.2 ± 1.6	9.2 ± 1.7
Kalmia latifolia	122.9 ± 28.3	73.0 ± 23.5	52.3 ± 11.2	14.0 ± 4.1
Viburnum acerifolium	1.5 ± 0.7	4.3 ± 1.8	10.4 ± 2.9	1.4 ± 0.9 **
Other shrub	5.2 ± 2.6	1.2 ± 0.9	2.5 ± 0.9	0.6 ± 0.3
Total shrub	129.6 ± 28.7	78.4 ± 23.8	65.2 ± 12.1	16.0 ± 4.1 *
Total understory	142.4 ± 28.4	92.1 ± 23.7	77.2 ± 12.1	25.8 ± 4.4 *
Acer rubrum	8.1 ± 4.2	20.2 ± 13.6	29.1 ± 11.0	3.4 ± 1.6 *
Amelanchier arborea	15.6 ± 7.2	5.6 ± 3.3	23.5 ± 6.4	10.0 ± 5.7
Fagus grandifolia	307.1 ± 43.2	180.9 ± 30.9 **	461.2 ± 55.6	299.8 ± 46.9 **
Hamamelis virginiana	23.6 ± 7.2	34.7 ± 10.8	79.1 ± 19.1	16.8 ± 7.0 *
Quercus rubra	3.8 ± 1.5	5.5 ± 2.4	67.8 ± 18.5	17.1 ± 7.5 **
Other tree	67.0 ± 18.6	72.9 ± 26.0	71.0 ± 15.3	49.5 ± 14.0
Total tree	425.0 ± 48.6	328.8 ± 39.6 *	731.7 ± 66.0	401.1 ± 48.8 **
Frequency				
Total graminoids	0.16	0.24	0.47	0.45
Aralia nudicaulis	0.51	0.69	0.46	0.39
Dennstaedtia punctilobula	0.41	0.32	0.19	0.15
Gaultheria procumbens	0.67	0.61	0.86	0.65 **
Maianthemum canadense	0.41	0.68 **	0.68	0.71
Mitchella repens	0.47	0.53	0.51	0.55
Pteridium aquilinum	0.30	0.29	0.31	0.26
Trientalis borealis	0.67	0.78	0.74	0.79
Other forbs	0.79	0.81	0.94	0.85
Kalmia latifolia	0.48	0.51	0.54	0.42
Viburnum acerifolium	0.37	0.51	0.45	0.24 **
Other shrubs	0.26	0.22	0.35	0.14 **
Acer rubrum	0.82	0.75	0.78	0.80
Amelanchier arborea	0.36	0.37	0.56	0.47
Fagus grandifolia	0.88	0.80	0.88	0.88
Hamamelis virginiana	0.53	0.46	0.53	0.29 **
Quercus rubra	0.68	0.76	0.77	0.80
Other trees	0.77	0.61	0.77	0.50 **

Table 6 (continued)

	1-burn		2-burn	
	High ($n = 73$)	Low ($n = 59$)	High ($n = 93$)	Low ($n = 66$)
Importance value				
Total graminoids	0.01 ± 0.01	0.01 ± 0.01	0.05 ± 0.01	0.04 ± 0.02
Aralia nudicaulis	0.10 ± 0.03	0.21 ± 0.04 **	0.07 ± 0.02	0.10 ± 0.02
Dennstaedtia punctilobula	0.16 ± 0.04	0.10 ± 0.03 *	0.07 ± 0.02	0.08 ± 0.03
Gaultheria procumbens	0.03 ± 0.01	0.03 ± 0.01	0.03 ± 0.01	0.05 ± 0.01
Maianthemum canadense	0.02 ± 0.01	0.05 ± 0.01	0.03 ± 0.01	0.09 ± 0.02 **
Mitchella repens	0.02 ± 0.01	0.03 ± 0.01	0.02 ± 0.01	0.03 ± 0.01 **
Pteridium aquilinum	0.03 ± 0.01	0.03 ± 0.01	0.08 ± 0.02	0.08 ± 0.02
Trientalis borealis	0.03 ± 0.01	0.05 ± 0.01	0.05 ± 0.01	0.06 ± 0.01
Other forbs	0.14 ± 0.03	0.11 ± 0.02	0.10 ± 0.02	0.10 ± 0.02
Total forbs	0.54 ± 0.05	0.60 ± 0.05	0.46 ± 0.04	0.61 ± 0.04 *
Kalmia latifolia	0.36 ± 0.05	0.27 ± 0.05	0.35 ± 0.04	0.28 ± 0.05
Viburnum acerifolium	0.03 ± 0.02	0.10 ± 0.03	0.11 ± 0.03	0.05 ± 0.02
Other shrubs	0.05 ± 0.02	0.01 ± 0.01	0.04 ± 0.01	0.02 ± 0.01
Total shrubs	0.44 ± 0.05	0.39 ± 0.05	0.50 ± 0.04	0.35 ± 0.05 *
Acer rubrum	0.03 ± 0.01	0.09 ± 0.03 *	0.03 ± 0.01	0.02 ± 0.01
Amelanchier arborea	0.05 ± 0.02	0.02 ± 0.01	0.04 ± 0.01	0.04 ± 0.01
Fagus grandifolia	0.62 ± 0.05	0.54 ± 0.05	0.58 ± 0.02	0.65 ± 0.05
Hamamelis virginiana	0.11 ± 0.04	0.12 ± 0.03	0.16 ± 0.03	0.05 ± 0.02 *
Quercus rubra	0.04 ± 0.02	0.05 ± 0.02	0.09 ± 0.02	0.09 ± 0.03
Other trees	0.15 ± 0.03	0.18 ± 0.04	0.10 ± 0.02	0.14 ± 0.03

values of tree and shrub richness (number of species per 6 m^2 combined plot) compared with low-intensity zones (Table 5). Forb and graminoid richness did not differ between high- and low-intensity zones.

4.2.2. Cumulative height

Total tree cumulative height was 1.3- and 1.8-fold greater in the high-intensity zones than in low-intensity zones of the 1- and 2-burn areas (Table 6). Shrub cumulative height was 4 times greater in the high-intensity zones of the 2-burn areas, with no discernible differences between intensity zones in the 1-burn areas. Cumulative heights of forbs and graminoids did not differ between the high- and low-intensity zones. Individual species' cumulative height values for high- and low-intensity zones are listed in Table 6.

4.2.3. Species importance values

The *IV* of shrubs was 1.4 times higher in the high-intensity zones than in the low-intensity zones in the 2-burn areas. That for forbs was 0.75 times lower in the high-intensity zones (Table 6). The discriminant analysis successfully separated the treatment areas based upon *IV* data from the 1992 transect plots (Fig. 4). The first canonical axis accounted for 46% of the total variation in the data set, and had a canonical correlation of 0.41 ($P < 0.001$) with the grouping variable. This axis separated the 1- and 2-burn areas based upon greater *IV* of ***Pteridium aquilinum*** and graminoids in the 2-burn areas, and greater *IV* of ***Aralia nudicaulis*** in the 1-burn areas (respective within-group correlations: 0.41, 0.41, −0.43). Caution should be used in the interpretation of this axis since it is possible that the 1992 species *IV* in different fire frequency treatment areas were confounded by unequal pre-fire species distributions. Fire frequency effects were more appropriately analyzed in the previous section.

The comparison of interest is between the high- and low-intensity zones of each burn treatment. These zones separated along the second canonical axis, which was defined by greater ***Maianthemum canadense*** *IV* in the low intensity zones (within

Table 7
Status of 1992 tall *Quercus rubra* advance regeneration in: (A) fire frequency treatments at the Tompkins and Broome prescribed fire study sites, southern New York, USA; (B) fire intensity treatment zones at the Broome study site. Values presented are mean ± 1 SE densities (stems ha^{-1}) of *Q. rubra* and all tree species for: (A) stems over 1.4 m tall and less than 8.9 cm dbh; (B) stems over 1.0 m tall and less than 8.9 cm dbh. The sample size (n) for each treatment is provided

	Tall *Quercus rubra*	Total tall trees
(A) Fire frequency treatments at Tompkins and Broome		
Tompkins		
0-burn (n = 36)	0 ± 0	7990 ± 1160
1-burn (n = 78)	420 ± 170	9780 ± 1080
2-burn (n = 72)	590 ± 230	7150 ± 1010
Broome		
0-burn (n = 36)	0 ± 0	3610 ± 850
1-burn (n = 78)	40 ± 40	6480 ± 980
2-burn (n = 72)	40 ± 40	5000 ± 1380
(B) Fire intensity treatment at Broome		
1-burn		
High (n = 73)	100 ± 50	10230 ± 1030
Low (n = 59)	140 ± 70	8430 ± 910
2-burn		
High (n = 93)	1680 ± 480	19130 ± 1650
Low (n = 66)	360 ± 170	11260 ± 1330

group correlation: 0.45). This axis accounted for 36% of the total variation, with a canonical correlation of 0.37 ($P < 0.001$). These results are consistent with the ANOVA results, which showed *M. canadense* had greater *IV* in the low-intensity zones of both treatment areas (the difference only being significant for the 2-burn areas). ANOVA detected additional *IV* differences between high- and low-intensity zones for *D. punctilobula, Aralia nudicaulis, Acer rubrum*, and *H. virginiana* (Table 6). Of these species, only *Aralia nudicaulis* exhibited consistent fire intensity responses in both treatment areas.

4.3. Status of Q. rubra regeneration

Table 7(A) presents the mean density of tall (over 1.4 m high) *Q. rubra* advance regeneration in each fire treatment area in 1992. These data show that there were greater densities of tall *Q. rubra* regeneration in the burned areas of both study sites, but the species never accounted for more than 8% and 1% of the tall regeneration at Tompkins and Broome, respectively. There were no tall (over 1.4 m) *Q. rubra* sampled on the 4 m^2 plots in the unburned areas of either site. Table 7(B) lists the mean densities of tall (over 1.0 m high) *Q. rubra* advance regeneration in the different fire intensity zones of Broome in 1992. *Quercus rubra* accounted for less than 3% of the tall advance regeneration in all treatments except the twice-burned, high-intensity zones, where it accounted for about 9% of all tall stems. In this treatment, there were over 1600 tall (over 1.0 m) *Q. rubra* stems per hectare.

5. Discussion

5.1. Fire effects on species richness

Contrasting trends in species richness following fire have been reported. Some studies have shown increases immediately following fire (Lewis and Harshbarger, 1976; Gilliam and Christensen, 1986; White et al., 1991) while others report decreases (Rego et al., 1991). Regardless, available evidence suggests that richness tends to return to predisturbance levels within 5–15 years after a fire (Gilliam and Christensen, 1986; Rego et al., 1991). In this study richness was reported at two scales: by the number of species sampled in treatment areas, and by the number of species per 4 m^2 plot. The data showed little change in the total number of tree and shrub species sampled in all treatment areas at the two study sites (Table 2). At Broome, forb richness changed little in all treatment areas, but at Tompkins richness increased in the 0-burn and 2-burn areas. Field observations revealed that many of the colonizing species at Tompkins occurred infrequently and on only a few plots traversed by skid trails. This fact, in conjunction with the inconsistent responses of richness in relation to fire, suggests that factors other than fire (e.g. soil scarification and canopy thinning) influenced study site richness at Tompkins. Considered on a per plot basis, significant increases in forb richness occurred in the burned areas at both sites. It appears that this increase is due primarily to increased frequencies of species common to the sites prior to the fires (see species frequencies, Table 4), rather than widespread establishment of colonizing species. These trends suggest that the fires did not create widespread microsites favorable for colonization by new species, but did stimulate the clonal expansion of existing species. Adams (1984) reported abundant vegetative sprouting of perennial forbs during the first post-fire growing season.

5.2. Fire effects on tree regeneration

In the Tompkins burned areas, tree regeneration (stems less than 8.9 cm dbh) density decreased between 1979 and 1992 owing to reduced densities of all species except *Amelanchier arborea, F. grandifolia*, and *H. virginiana*. However, there was no accompanying decrease in total tree cumulative height (Table 4). Fewer trees in 1992 having cumulative heights comparable to the 1979 levels indicates taller average stem heights in 1992, and likely reflects self-thinning among the even-aged regeneration. Self-thinning may also have caused the decreased tree richness in all treatment areas at Tompkins (Table 3). In fact, the decline in *Q. rubra* frequency in all treatment areas probably contributed greatly to the lower richness values.

Most tree regeneration at both study sites were sprouts of *F. grandifolia*, which often formed dense thickets. Sprouting may have been stimulated by fire injury to roots, or fire-induced injury/mortality to parent trees. *Fagus grandifolia* has been demonstrated to sprout following root injury (Jones and Raynal, 1988), cutting of mature trees (Jones et al., 1989), and stress related to beech bark disease (Ostrofsky and McCormack, 1986). Ward (1961) and Held (1983) found most sprouts on lateral roots at or near the ground surface. Even though beech bark symptoms were evident at both study sites, additional *F. grandifolia* sprouting may have been initiated following injury to shallow or exposed roots, and/or from fire-induced stress among overstory trees.

At Broome, the cumulative height of tree regeneration was greater in the high-intensity zones compared with the low-intensity zones. Also, the high-intensity zones of the twice-burned areas had greater species richness than the low-intensity zones. The high-intensity fires resulted in heavier mortality to the upper canopy and subcanopy trees (Nyland et al., 1982). This would have resulted in greater understory light levels, and may explain the greater cumulative heights in the high-intensity zones and 1980 burn area. High-intensity fires may also have induced more sprouting (at least for *F. grandifolia*), and created better germination sites by a greater consumption of the litter layer.

5.3. Fire effects on understory species

At Tompkins, the cumulative height of forbs and shrubs in 1992 was greater than that for 1979 in the 2-burn areas. This was not the case for the 0- or 1-burn areas. The inconsistency between burn areas reflects the formation of a tight tree regeneration canopy approximately 3–4 m high in the 1-burn areas by 1992. Shading by these trees probably adversely affected two important, relatively shade-intolerant understory species, *Rubus* spp. and *Dennstaedtia punctilobula*. Together, these species accounted for 52% of the 1992 total understory cumulative height in the 2-burn areas, but only 23% in the 1-burn areas, where they were overtopped (Table 4).

At Broome, the 1992 forb cumulative height was significantly greater than 1979 levels in both burned areas, but that for shrubs did not change (Table 4). Shramek (1985) reported that *K. latifolia* and *V. acerifolium* stems were generally less than 30 cm tall in 1983. Apparently, the continued short heights of shrubs in 1992 reflects relatively slow recovery rates for these species. Rego et al. (1991) also reported slow shrub recovery after fire, and Hooper (1969) cited the effectiveness of prescribed fire in reducing *K. latifolia* cover for several years. The greater cumulative heights of shrubs in the high-intensity zones at Broome may reflect more favorable growing conditions due to overstory mortality.

No perennial forb species were adversely affected by the fires. In fact, Adams (1984) reported that most met or surpassed their pre-burn densities the first growing season after the fires. However, individual species responses to fire frequency varied between the two study sites. This may be due to confounding interactions of the 1976 canopy thinning at Tompkins. At Broome, the cover of graminoids and forbs were unaffected by fire intensity, but owing to more vigorous shrub growth in the high-intensity zones, overall forb importance values were lower there. This trend was present in both burn areas, but significant only in the 2-burn areas.

5.4. Fire effects on overall community composition

Past research has shown that fire frequency and fire intensity both importantly affect community composition. Lewis and Harshbarger (1976), White et al. (1991), and DeSelm and Clebsch (1991) all demonstrated that several years of repeated annual or biennial fires are necessary to substantially alter community composition. Heinselman (1981) and Rice (1993) showed that fires of varying intensity are important factors in creating community diversity at both the landscape and microtopographic scales.

In this study, discriminant analyses and ANOVA identified significant changes in species composition between 1979 and 1992 following fire at both study sites (Figs. 2 and 3, Table 4). In general, fire frequency treatments appeared to consistently affect only two tree species. *Quercus rubra IV* decreased in all treatment areas at Tompkins, and received no apparent benefit from the fire treatments, and at Broome, *Q. rubra IV* declined on burned areas. *Fagus grandifolia IV* generally appeared to increase following fires. The fires also appeared to improve the status of forbs relative to shrubs. In general, the importance values of *V. acerifolium* and *K. latifolia* decreased on burned areas. However, *Rubus* spp., which were abundant at Tompkins following the 1976 shelterwood cuts were able to persist through 1992 in the 2-burn area only. Species composition of the high- and low-intensity zones at Broome did not show consistent, significant differences.

In general, in the 8–12 years following these experimental fires, evident structural differences developed between the various fire frequency and intensity areas. Compositional changes were minor, however. These reflected small, but significant, changes in the importance values of a few common species. Apparently, the frequencies of these fires and the intensities at which they burned were insuffi-

Table 8

Summary of common understory perennial species responses to fire frequency in relation to rhizome depth and growth rates at the Tompkins and Broome prescribed fire study sites, southern New York, USA. Symbols represent significant positive changes (+), no change (·), and missing data (○) in frequency and cumulative height between 1979 and 1992 within the 0-, 1-, and 2-burn areas. Symbols are presented in pairs. The left symbol of each pair represents the species' response at Tompkins, the right at Broome. A space indicates that the species was not important at the study site

Species	Frequency			Cumulative height			Rhizome depth [a]	Rhizome growth rate [b] ($cm\ year^{-1}$)
	0-burn	1-burn	2-burn	0-burn	1-burn	2-burn		
Gaultheria procumbens	· ·	· ·	· ·	· ·	· +	· +	l	> 10 max
Maianthemum canadense	· ·	+ +	· +	○○	○○	○○	l	15–30
Mitchella repens	· ·	· ·	· ·	+ ·	· +	· +	l	< 10 max
Trientalis borealis	· ·	· +	· +	· ·	· +	+ +	l	80 max
Uvularia sessilifolia	· ·	· ·	· +	· ·	· ·	+ +	h	?
Dennstaedtia punctilobula	·	·	+	·	·	+	h	?
Rhododendron periclymenoides	·	·	·	·	·	·	h [c]	?
Rubus spp.	·	·	·	·	·	+	h–m [c]	?
Vaccinium spp.	·	·	·	·	·	·	m	?
Viburnum acerifolium	· ·	· ·	· ·	· ·	· ·	· ·	h–m [c]	?
Aralia nudicaulis	·	·	·	·	·	·	m	25–80
Kalmia latifolia	·	·	·	·	·	·	None [d]	–

[a] Rhizome depths from Flinn and Wein (1977): m, mineral soil; h, humus; l, litter.
[b] Rhizome growth rates from Sobey and Barkhouse (1977).
[c] Rhizome depth based upon personal inspection.
[d] *Kalmia latifolia* does not expand rhizomatously (Anderson and Egler, 1988).

cient to substantially change community composition.

Other studies (e.g. Abrahamson, 1984; Rego et al., 1991; Schmalzer and Hinkle, 1992; Matlack et al., 1993) have attributed community stability following fire to: (1) sprouting of dominant species; (2) failure of subordinate species to increase in numbers and/or size; (3) failure of invasive species to persist. This study yielded consistent results: (1) all dominant species resprouted; (2) with the exception of *D. punctilobula* at Tompkins, no subordinate species became important components (attaining *IV* > 0.05) of the two communities following fire; (3) no ruderal species became important and persisted through 1992.

5.5. *Competitive responses of understory species following fire*

A number of studies have shown that community recovery following fire depends primarily upon sprouting, and that initial floristic composition greatly influences post-disturbance composition (Schmalzer and Hinkle, 1992; Matlack et al., 1993). It has been further suggested that post-fire competitiveness among sprouting species is related to interspecific variability in below-ground morphologic traits (e.g. depth of perennating tissues, root depth, and rhizome growth rates). Flinn and Wein (1977) proposed that species with perennating tissue in the litter layer would not likely survive even light fires, but those with buds in the mineral soil are suited for avoiding intense fires. Chapin and Van Cleve (1981) argued that shallow rooted species, if capable of surviving fires, are better able to compete for nutrient pools in the upper soil horizons, compared with deeply rooted species. Sobey and Barkhouse (1977) and Antos and Zobel (1984) classified forest herbs based upon rhizome growth rates, suggesting that persistent species with slow growth rates are 'tolerant', and those with faster growth rates are 'competitive', based upon Grime's (1974) competitive strategy model.

Table 8 summarizes the significant changes in frequency and cumulative height for the dominant perennial forb and shrub species at the two study sites, making reference to the presence/absence of rhizomes, rhizome depths and rhizome growth rates. *Kalmia latifolia* was the only non-rhizomatous species (Anderson and Egler, 1988) at the two study sites. Although it sprouted readily, it did not change significantly in frequency or cumulative height. The group of perennial forbs having rhizomes in the litter or humus not only survived the fires, they increased more in frequency and cover than species having rhizomes in the mineral soil. Nyland et al. (1982) reported increased post-fire nutrient content in the humus and compacted leaf litter, but not in the mineral soil after the 1980 fires. Perhaps species having rhizomes in the litter and humus produced more adventitious root masses where post-fire nutrient accumulations occurred, enabling greater competitiveness for the new resource base. The rhizomes of *Aralia nudicaulis* are located within the mineral soil. So even though it has one of the highest rhizome growth rates of the common species in this study (Sobey and Barkhouse 1977), *Aralia nudicaulis* may not have been competitive for the new nutrient pool. Although determining relationships between morphology and post-fire competitiveness was not a specific objective of this study, our observations were consistent with Chapin and Van Cleve's hypothesis (1981), but did not reflect the predictions of Flinn and Wein (1977).

5.6. *Status of* Q. rubra *regeneration*

Adams (1984) reported that *Q. rubra* increased more in density and frequency than all other tree species during the first growing season following the 1980 fires. It was the most important of regenerating species at both study sites at that time. But by 1992, importance values for *Q. rubra* had decreased in all treatment areas at Tompkins, and in the burned areas at Broome. This suggests no long-term benefit for *Q. rubra* from fire at Tompkins, and an adverse affect of fire at Broome, owing to superior growth of competing species. Sander et al. (1976) suggested that approximately 1100 well-distributed stems of *Q. rubra* per hectare, at least 1.4 m tall, are required for this species to remain codominant when a stand is regenerated. The 1992 permanent plot survey of *Q. rubra* indicates that the use of prescribed fire, with or without overstory thinning, did not increase tall oak regeneration sufficiently to satisfy these criteria. However, preliminary data from the Broome fire intensity plots suggest that tall *Q. rubra* may occur at the required densities in the high-intensity zones of the twice-burned areas. These results should be considered with caution since: (1) they include densities of stems over 1 m, not over 1.4 m; (2) competing species occur in the taller height classes in equal or greater numbers, as illustrated by the lack of significant improvement in *Q. rubra* importance values in the high intensity zones.

6. Conclusions

This study showed that relatively infrequent (0, 1 or 2 fires over 12 years), low-intensity, prescribed fires in two transition oak stands resulted in only minor changes to understory and tree regeneration species compositions. Forb species richness increased, on the scale of 4 m^2 plots, in burned areas. Changes in richness were due primarily to increased frequencies of pre-existing species, rather than the establishment and persistence of post-fire colonizers. Forb cumulative height increased in the 8–12 years following fire at both study sites. With the exception of *Rubus* spp. at Tompkins, the 8–12 year post-burn shrub cumulative heights did not differ from pre-burn values, reflecting slow shrub recovery rates. At Broome, areas receiving relatively greater intensity fires had greater cumulative heights of trees and shrubs, but no consistent differences in understory or tree regeneration species compositions. Observed relationships between species responses, rhizome depths, and the locations of post-fire nutrient accumulations suggest the importance of below-ground morphologic traits in affecting post-fire competitiveness. In general, the status of *Quercus rubra* among competing tree regeneration species (particularly *F. grandifolia*) was not improved by the use of prescribed fire, or fire in conjunction with shelterwood seed cutting. After 8–12 years, tall (stems over 1.4 m) *Q. rubra* is not present at either site at densities sufficient to justify canopy removal according to the criteria of Sander et al. (1976). *Quercus rubra* regeneration over 1 m tall, but not necessarily over 1.4 m tall, may be present in the recommended densities in the high-intensity zones at Broome. However, the importance value of *Q. rubra* was no greater in these areas, reflecting the abundance of competing species of equal or greater heights.

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